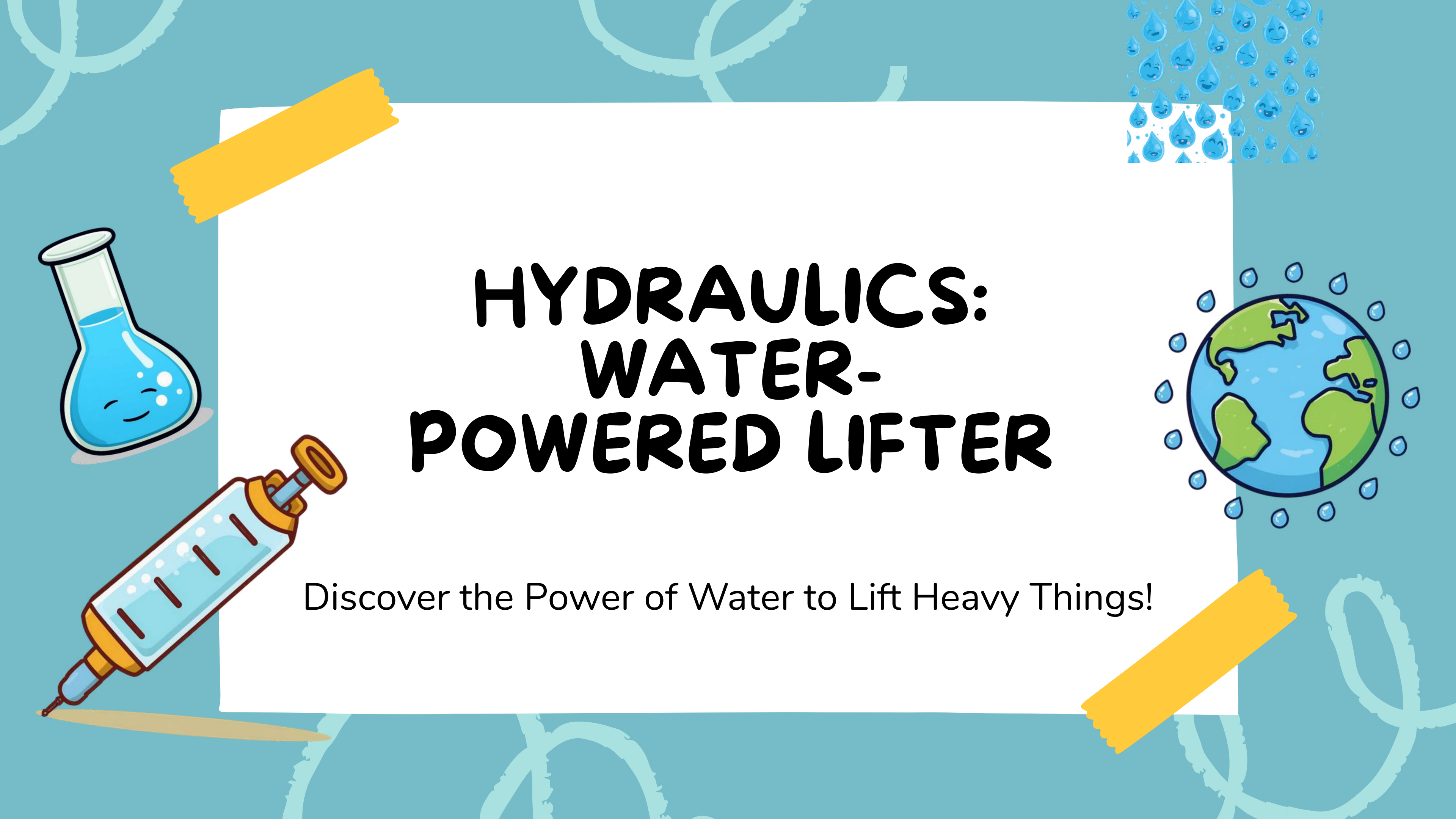




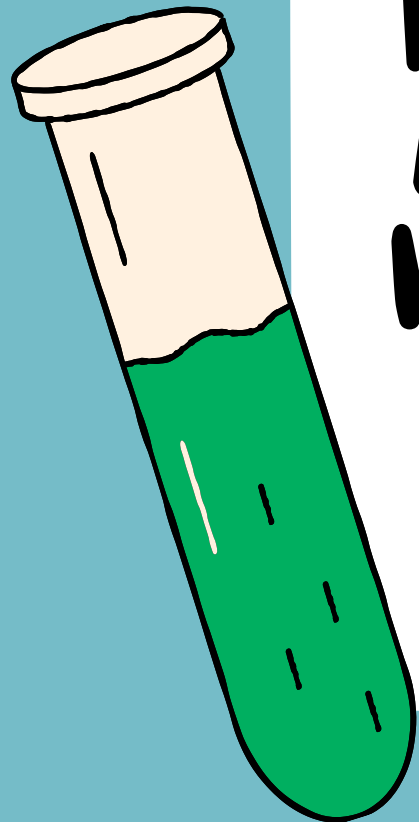
HYDRAULICS: WATER- POWERED LIFTER

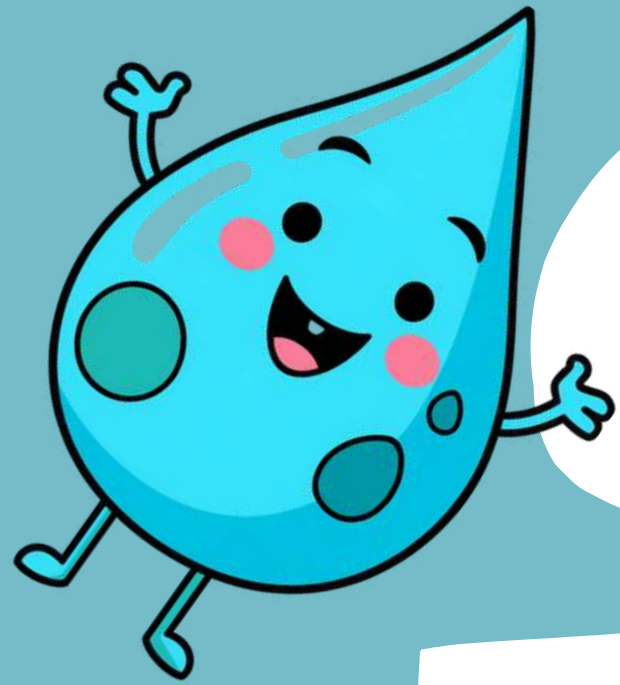
Discover the Power of Water to Lift Heavy Things!





How does this big machine lift such heavy things with water?





What is Hydraulics?

Definition

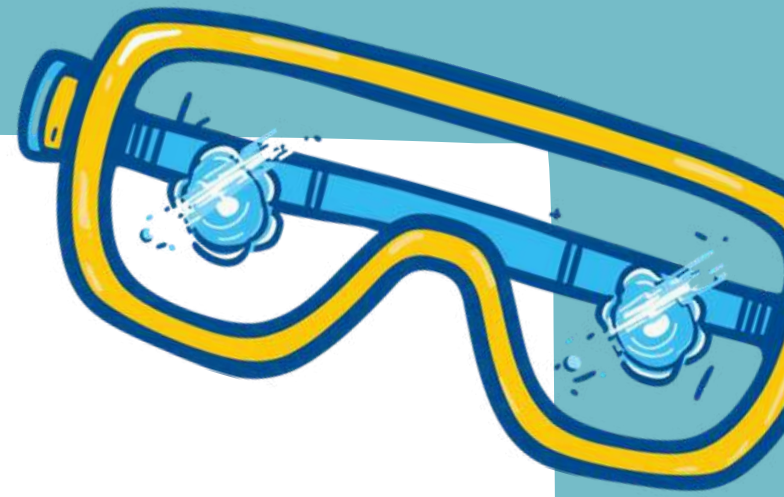
Hydraulics means using water or other liquids to move or lift things.

Key Idea

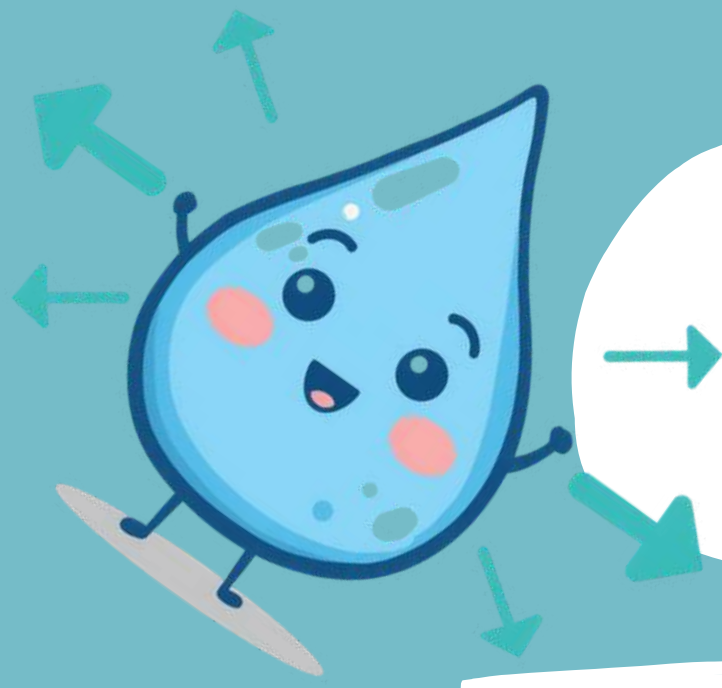
Water helps us do big jobs with small pushes!



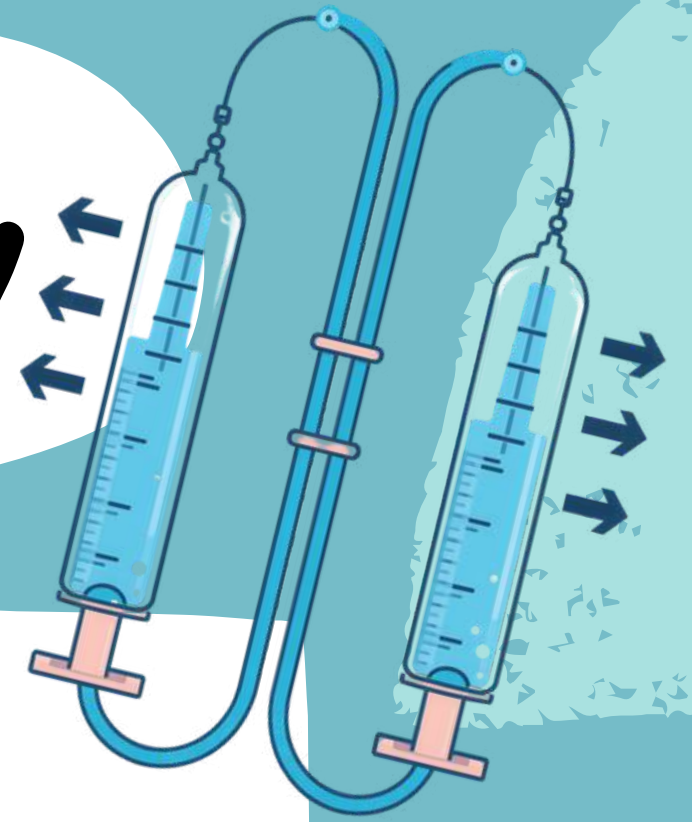
Water vs. Air: The Science of Squish



A liquid cannot be squished into a smaller space — it always takes up the same amount of room, no matter how hard you push it! This is called incompressibility.



Pascal's Law Made Easy



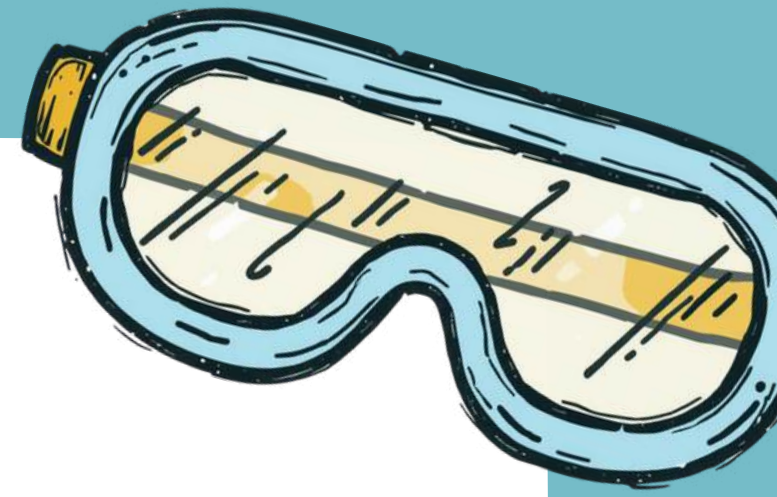
The Rule

When you push water in a closed pipe, the push spreads everywhere equally!

Example

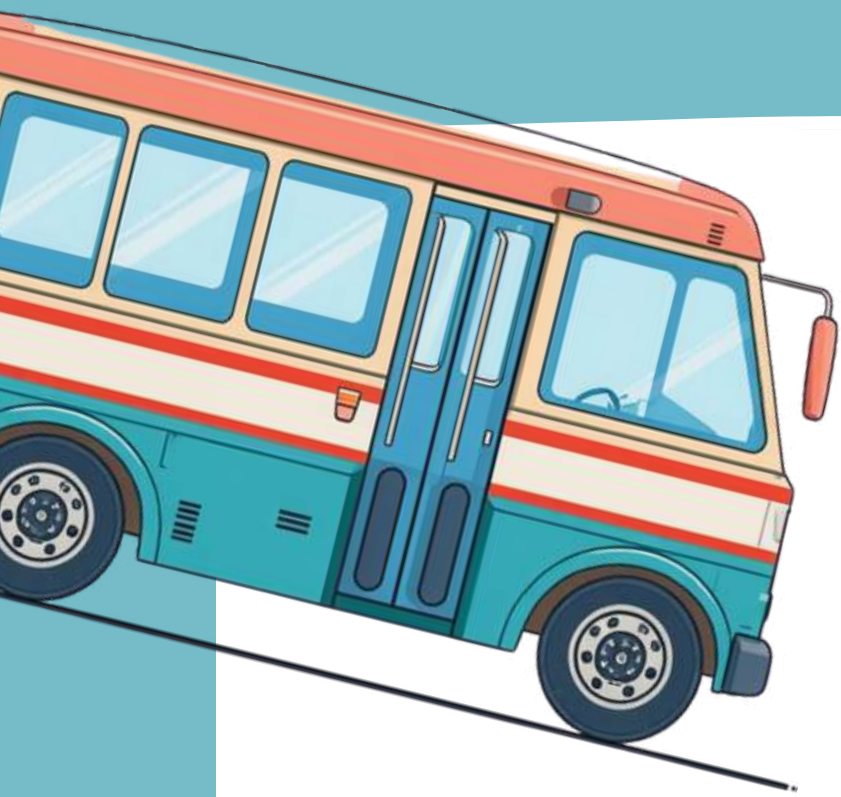
If you press one side of a water pipe, the other side feels the same push!

Push and Pull with Syringes

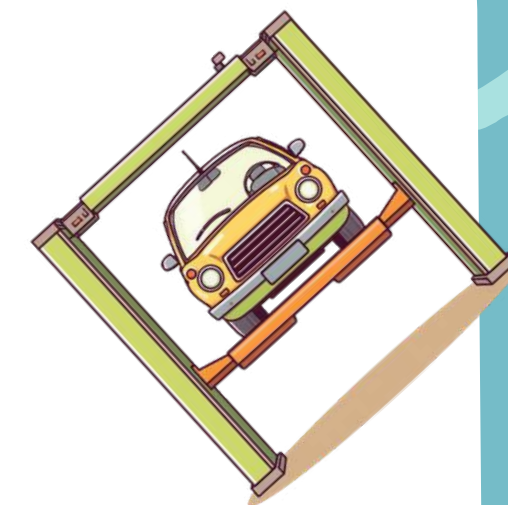


Push one syringe, and the other pushes out too. This is how we lift things!





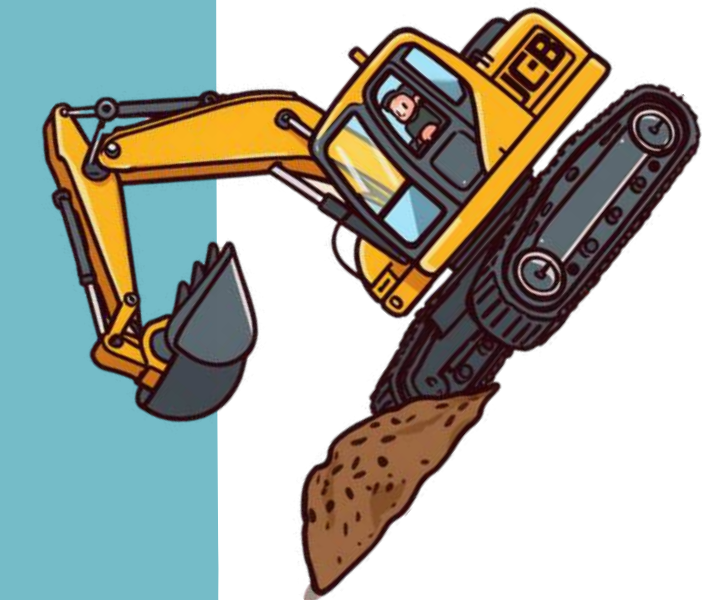
Hydraulics in India Around Us



JCB at Construction Sites

Hydraulic Brakes on Buses

Car Lifting Platforms



JCB machines use water power to lift heavy rocks and dig soil at building sites!

Buses and trains use hydraulic brakes to stop safely. Water pressure helps slow down heavy vehicles!



Let's Build Our Water-Powered Lifter!

MATERIALS NEEDED:

- 2 Plastic Syringes (different sizes if possible)
- 1 Flexible Plastic Tube (IV set tube works great)
- Water (add a drop of Neel/Ink for color!)
- Cardboard base and a small platform
- Cello tape or Glue



LET'S BUILD OUR WATER- POWERED LIFTER:

Step 1: Fill one syringe completely with colored water, making sure there are no air bubbles

Step 2: Connect the tube to the tip of the water-filled syringe. Push slowly until water reaches the very end of the tube.

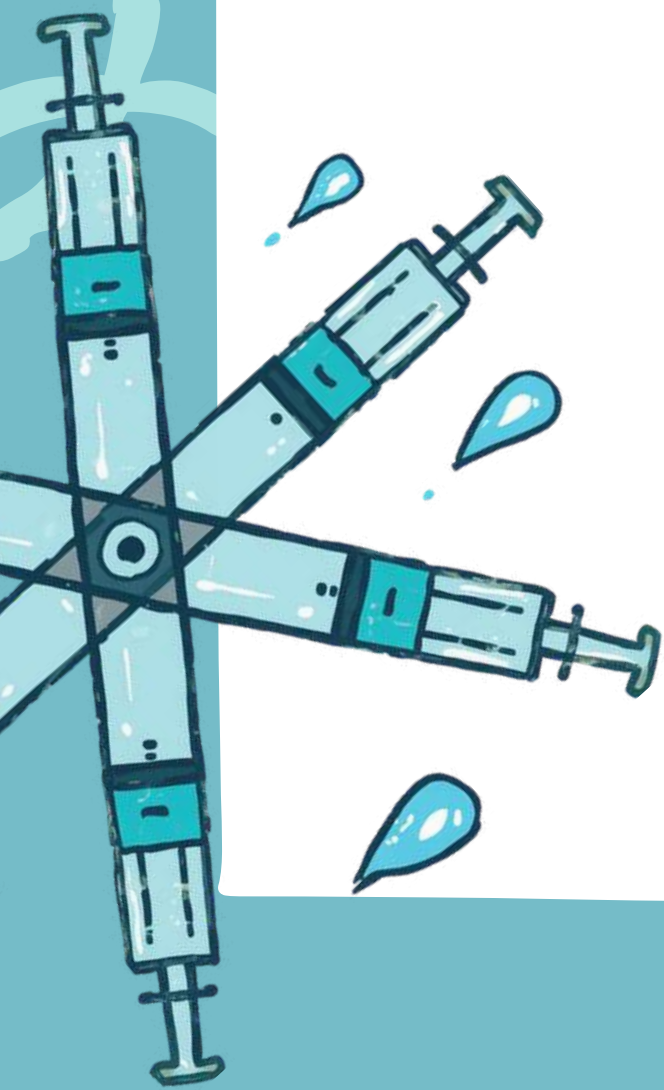
Step 3: Connect the other end of the tube to the empty syringe. Ensure the connection is tight.

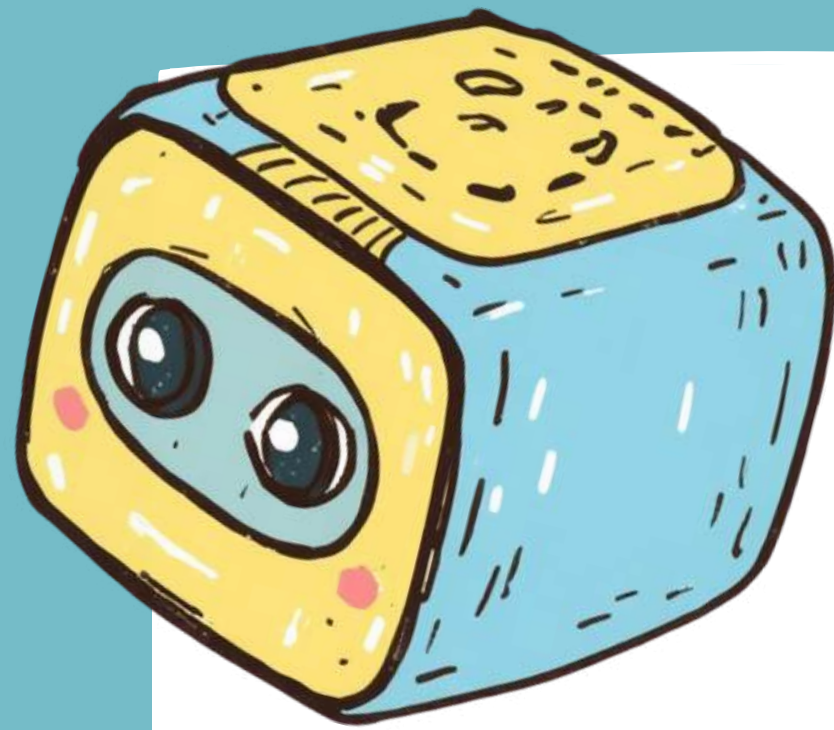
Step 4: Secure the syringes to your cardboard frame. One will be your "Controller" and the other will be the "Lifter" under your platform.

Watch the Lifter in Action



Look how
water lifts the
platform when
we push here!





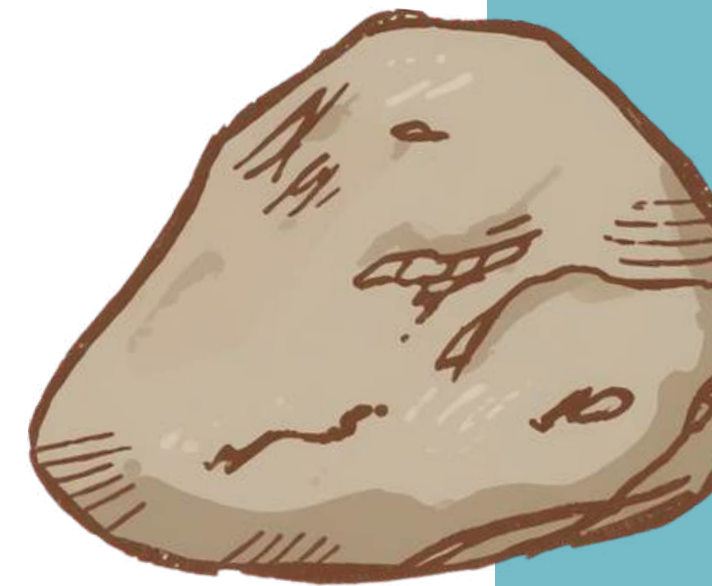
Challenge Time!



Eraser (Light)

Sharpener (Medium)

Heavy Stone (Hard)



Can your lifter pick up these objects? Try each one and see what happens!



Experiment and note which objects your hydraulic lifter can pick up. What works best?

What Did We Learn?



Water cannot
be squished -
it's
incompressible!



Push one syringe,
water moves
everywhere
equally!



Small push can
lift heavy
objects with
hydraulics!





Ready to be a Hydraulic Hero?

Let's see how much you know! Answer these fun questions and show off your water power skills. You're becoming experts in the power of water!



Questions & Fun Facts!

Did you know? The hydraulic lift can lift cars weighing thousands of kilos! Ask your questions now!





Thank You & Keep Exploring!

THANK YOU FOR JOINING THE WATER-POWERED LIFTER ADVENTURE!
KEEP EXPERIMENTING AND ASKING QUESTIONS. - PLANRS ACADEMY